**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 150 · Paper 2: Writing Code from a Scenario

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What must be true before binary search is used?

Q2. In OCR ERL, what is 2 ^ 3?
A. 6
B. 8
C. 9
D. 23

Q3. “OR is false whenever both inputs are true.” Correct this. Explain why.

Q4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Identify input — List all inputs, types and relevant constraints before writing code.

Design — State algorithm steps separately from program syntax.

Select and iterate — Write appropriate IF/ELSE and loops rather than unnecessary repetition.

Output and test — Check both a typical value and at least one boundary/unexpected condition.

## New learning

Identify input: List all inputs, types and relevant constraints before writing code.

Design: State algorithm steps separately from program syntax.

Select and iterate: Write appropriate IF/ELSE and loops rather than unnecessary repetition.

Output and test: Check both a typical value and at least one boundary/unexpected condition.

## Worked example

Write an algorithm to input three integers and output their total.

1. Set total=0.

2. Repeat three times to input a number.

3. Add each entered number to total.

4. Output total after the loop.

# We Do · Practise with guidance

### Guided 1

Design a program to input five marks and count those ≥ 50.

Worked method: Print passCount after all entries.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Cast the typed file size to an integer before comparing it with 100.

```
sizeText = input("File size in MB?")
sizeMB = ...
if sizeMB > 100 then
    print("Large")
else
    print("Small")
endif
```

write

### Question 1

Write OCR-style pseudocode to output the larger of two input values. [4 marks]

### Question 2

Explain why an accumulator starts at zero when summing values. [2 marks]

### Question 3

State why a final output statement should be outside a loop when only one total is required. [2 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Exit Ticket · Answer before checking

The same program line must be copied five times instead of using a loop.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. The list must be sorted. This lets the search safely rule out half the remaining items.

Q2. B — 8

Q3. When at least one input is true, including when both are true.

Q4. Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly.

### I Do / We Do pages · Guided practice

Q1. Print passCount after all entries.

### You Do pages · Code First

```
sizeText = input("File size in MB?")
sizeMB = int(sizeText)
if sizeMB > 100 then
    print("Large")
else
    print("Small")
endif
```

Expected result: Large (if input is "120")

### You Do pages · Independent questions

Q1. Input a,b; if a>b print(a), else print(b); equality may output either as both are largest.

Q2. Zero is the additive identity and avoids introducing an unwanted starting total.

Q3. Inside the loop would print intermediate totals repeatedly.

### Exit Ticket · Model answer

False

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

## 1. Copy the whole original question:

## 2. Write a complete new answer in GREEN PEN. Show working if needed: